neural learning about edges and corners: intro to convolutional neural networks

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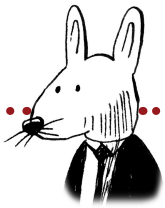


In this chapter

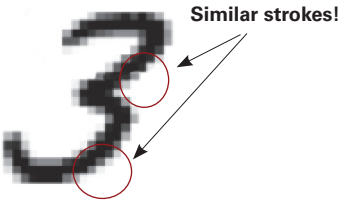
- Reusing weights in multiple places
- The convolutional layer

> The pooling operation used in convolutional neural networks is a big mistake, and the fact that it works so well is a disaster.
>
> —Geoffrey Hinton, from "Ask Me Anything" on Reddit

Reusing weights in multiple places

If you need to detect the same feature in multiple places, use the same weights!

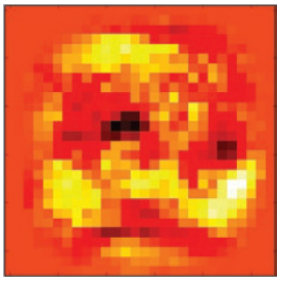
The greatest challenge in neural networks is that of overfitting, when a neural network memorizes a dataset instead of learning useful abstractions that generalize to unseen data. In other words, the neural network learns to predict based on noise in the dataset as opposed to relying on the fundamental signal (remember the analogy about a fork embedded in clay?).



Overfitting is often caused by having more parameters than necessary to learn a specific dataset. In this case, the network has so many parameters that it can memorize every fine-grained detail in the training dataset (neural network: "Ah. I see we have image number 363 again. This was the number 2.") instead of learning high-level abstractions (neural network: "Hmm, it's got a swooping top, a swirl at the bottom left, and a tail on the right; it must be a 2."). When neural networks have lots of parameters but not very many training examples, overfitting is difficult to avoid.

We covered this topic extensively in chapter 8, when we looked at regularization as a means of countering overfitting. But regularization isn't the only technique (or even the most ideal technique) to prevent overfitting.

As I mentioned, overfitting is concerned with the ratio between the number of weights in the model and the number of datapoints it has to learn those weights. Thus, there's a better method to counter overfitting. When possible, it's preferable to use something loosely defined as *structure*.



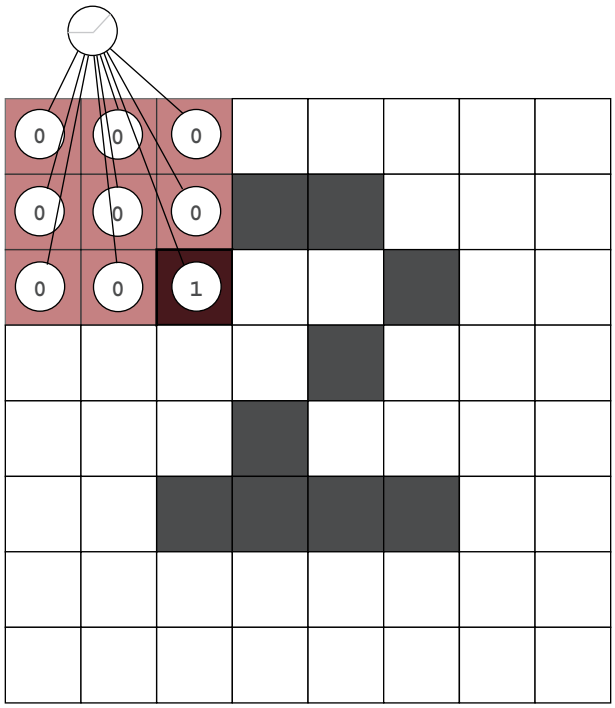
Structure is when you selectively choose to reuse weights for multiple purposes in a neural network because we believe the same pattern needs to be detected in multiple places. As you'll see, this can significantly reduce overfitting and lead to much more accurate models, because it reduces the weight-to-data ratio.

But whereas normally removing parameters makes the model less expressive (less able to learn patterns), if you're clever in where you reuse weights, the model can be equally expressive but more robust to overfitting. Perhaps surprisingly, this technique also tends to make the model smaller (because there are fewer actual parameters to store). The most famous and widely used structure in neural networks is called a *convolution*, and when used as a layer it's called a *convolutional layer*.

The convolutional layer

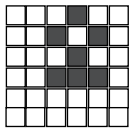
Lots of very small linear layers are reused in every position, instead of a single big one.

The core idea behind a convolutional layer is that instead of having a large, dense linear layer with a connection from every input to every output, you instead have lots of very small linear layers, usually with fewer than 25 inputs and a single output, which you use in every input position. Each mini-layer is called a convolutional *kernel*, but it's really nothing more than a baby linear layer with a small number of inputs and a single output.

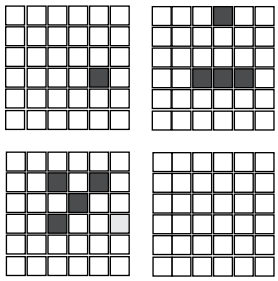


Shown here is a single 3 × 3 convolutional kernel. It will predict in its current location, move one pixel to the right, then predict again, move another pixel to the right, and so on. Once it has scanned all the way across the image, it will move down a single pixel and scan back to the left, repeating until it has made a prediction in every possible position within the image. The result will be a smaller square of kernel predictions, which are used as input to the next layer. Convolutional layers usually have many kernels.

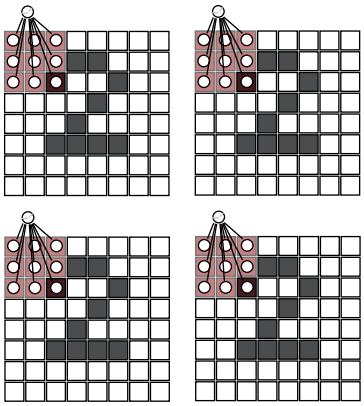
At bottom-right are four different convolutional kernels processing the same 8 × 8 image of a 2. Each kernel results in a 6 × 6 prediction matrix. The result of the convolutional layer with four 3 × 3 kernels is four 6 × 6 prediction matrices. You can either sum these matrices elementwise (sum pooling), take the mean elementwise (mean pooling), or compute the elementwise maximum value (max pooling).



The max value of each kernel's output forms a meaningful representation and is passed to the next layer.



Outputs from each of the four kernels in each position



Four convolutional kernels predicting over the same 2

The last version turns out to be the most popular: for each position, look into each of the four kernel's outputs, find the max, and copy it into a final 6 × 6 matrix as pictured at upper-right of this page. This final matrix (and only this matrix) is then forward propagated into the next layers.

There are a few things to notice in these figures. First, the bottom-right kernel forward propagates a 1 only if it's focused on a horizontal line segment. The bottom-left kernel forward propagates a 1 only if it's focused on a diagonal line pointing upward and to the right. Finally, the bottom-right kernel didn't identify any patterns that it was trained to predict.

It's important to realize that this technique allows each kernel to learn a particular pattern and then search for the existence of that pattern somewhere in the image. A single, small set of weights can train over a much larger set of training examples, because even though the dataset hasn't changed, each mini-kernel is forward propagated multiple times on multiple segments of data, thus changing the ratio of weights to datapoints on which those weights are being trained. This has a powerful impact on the network, drastically reducing its ability to overfit to training data and increasing its ability to generalize.

A simple implementation in NumPy

Just think mini-linear layers, and you already know what you need to know.

Let's start with forward propagation. This method shows how to select a subregion in a batch of images in NumPy. Note that it selects the same subregion for the entire batch:

```
def get_image_section(layer,row_from, row_to, col_from, col_to):
    sub_section = layer[:,row_from:row_to,col_from:col_to]
    return subsection.reshape(-1,1,row_to-row_from, col_to-col_from)
```

Now, let's see how this method is used. Because it selects a subsection of a batch of input images, you need to call it multiple times (on every location within the image). Such a `for` loop looks something like this:

```
layer_0 = images[batch_start:batch_end]
layer_0 = layer_0.reshape(layer_0.shape[0],28,28)
layer_0.shape

sects = list()
for row_start in range(layer_0.shape[1]-kernel_rows):
    for col_start in range(layer_0.shape[2] - kernel_cols):
        sect = get_image_section(layer_0,
                                 row_start,
                                 row_start+kernel_rows,
                                 col_start,
                                 col_start+kernel_cols)
        sects.append(sect)

expanded_input = np.concatenate(sects,axis=1)
es = expanded_input.shape
flattened_input = expanded_input.reshape(es[0]*es[1],-1)
```

In this code, `layer_0` is a batch of images 28 × 28 in shape. The `for` loop iterates through every (`kernel_rows` × `kernel_cols`) subregion in the images and puts them into a list called `sects`. This list of sections is then concatenated and reshaped in a peculiar way.

Pretend (for now) that each individual subregion is its own image. Thus, if you had a batch size of 8 images, and 100 subregions per image, you'd pretend it was a batch size of 800 smaller images. Forward propagating them through a linear layer with one output neuron is the same as predicting that linear layer over every subregion in every batch (pause and make sure you get this).

If you instead forward propagate using a linear layer with *n* output neurons, it will generate the outputs that are the same as predicting *n* linear layers (kernels) in every input position of the image. You do it this way because it makes the code both simpler and faster:

```
kernels = np.random.random((kernel_rows*kernel_cols,num_kernels))
                ...
kernel_output = flattened_input.dot(kernels)
```

The following listing shows the entire NumPy implementation:

```
import numpy as np, sys
np.random.seed(1)

from keras.datasets import mnist

(x_train, y_train), (x_test, y_test) = mnist.load_data()

images, labels = (x_train[0:1000].reshape(1000,28*28) / 255,
                  y_train[0:1000])


one_hot_labels = np.zeros((len(labels),10))
for i,l in enumerate(labels):
    one_hot_labels[i][l] = 1
labels = one_hot_labels

test_images = x_test.reshape(len(x_test),28*28) / 255
test_labels = np.zeros((len(y_test),10))
for i,l in enumerate(y_test):
    test_labels[i][l] = 1

def tanh(x):
    return np.tanh(x)

def tanh2deriv(output):
    return 1 - (output ** 2)

def softmax(x):
    temp = np.exp(x)
    return temp / np.sum(temp, axis=1, keepdims=True)

alpha, iterations = (2, 300)
pixels_per_image, num_labels = (784, 10)
batch_size = 128

input_rows = 28
input_cols = 28

kernel_rows = 3
kernel_cols = 3
num_kernels = 16

hidden_size = ((input_rows - kernel_rows) *
               (input_cols - kernel_cols)) * num_kernels

kernels = 0.02*np.random.random((kernel_rows*kernel_cols,
                                 num_kernels))-0.01

weights_1_2 = 0.2*np.random.random((hidden_size,
                                    num_labels)) - 0.1

def get_image_section(layer,row_from, row_to, col_from, col_to):
    section = layer[:,row_from:row_to,col_from:col_to]
    return section.reshape(-1,1,row_to-row_from, col_to-col_from)
```

```
for j in range(iterations):
    correct_cnt = 0
    for i in range(int(len(images) / batch_size)):
        batch_start, batch_end=((i * batch_size),((i+1)*batch_size))
        layer_0 = images[batch_start:batch_end]
        layer_0 = layer_0.reshape(layer_0.shape[0],28,28)
        layer_0.shape

        sects = list()
        for row_start in range(layer_0.shape[1]-kernel_rows):
            for col_start in range(layer_0.shape[2] - kernel_cols):
                sect = get_image_section(layer_0,
                                         row_start,
                                         row_start+kernel_rows,
                                         col_start,
                                         col_start+kernel_cols)
                sects.append(sect)

        expanded_input = np.concatenate(sects,axis=1)
        es = expanded_input.shape
        flattened_input = expanded_input.reshape(es[0]*es[1],-1)

        kernel_output = flattened_input.dot(kernels)
        layer_1 = tanh(kernel_output.reshape(es[0],-1))
        dropout_mask = np.random.randint(2,size=layer_1.shape)
        layer_1 *= dropout_mask * 2
        layer_2 = softmax(np.dot(layer_1,weights_1_2))

        for k in range(batch_size):
            labelset = labels[batch_start+k:batch_start+k+1]
            _inc = int(np.argmax(layer_2[k:k+1]) ==
                               np.argmax(labelset))
            correct_cnt += _inc

        layer_2_delta = (labels[batch_start:batch_end]-layer_2)\
                        / (batch_size * layer_2.shape[0])
        layer_1_delta = layer_2_delta.dot(weights_1_2.T) * \
                        tanh2deriv(layer_1)
        layer_1_delta *= dropout_mask
        weights_1_2 += alpha * layer_1.T.dot(layer_2_delta)
        l1d_reshape = layer_1_delta.reshape(kernel_output.shape)
        k_update = flattened_input.T.dot(l1d_reshape)
        kernels -= alpha * k_update

    test_correct_cnt = 0

    for i in range(len(test_images)):

        layer_0 = test_images[i:i+1]
        layer_0 = layer_0.reshape(layer_0.shape[0],28,28)
        layer_0.shape

        sects = list()
        for row_start in range(layer_0.shape[1]-kernel_rows):
            for col_start in range(layer_0.shape[2] - kernel_cols):
                sect = get_image_section(layer_0,
                                         row_start,
                                         row_start+kernel_rows,
```

```
                                        col_start,
                                        col_start+kernel_cols)
            sects.append(sect)

    expanded_input = np.concatenate(sects,axis=1)
    es = expanded_input.shape
    flattened_input = expanded_input.reshape(es[0]*es[1],-1)

    kernel_output = flattened_input.dot(kernels)
    layer_1 = tanh(kernel_output.reshape(es[0],-1))
    layer_2 = np.dot(layer_1,weights_1_2)

    test_correct_cnt += int(np.argmax(layer_2) ==
                            np.argmax(test_labels[i:i+1]))
if(j % 1 == 0):
    sys.stdout.write("\n"+ \
     "I:" + str(j) + \
     " Test-Acc:"+str(test_correct_cnt/float(len(test_images)))+\
     " Train-Acc:" + str(correct_cnt/float(len(images))))
```

```
I:0 Test-Acc:0.0288 Train-Acc:0.055
I:1 Test-Acc:0.0273 Train-Acc:0.037
I:2 Test-Acc:0.028 Train-Acc:0.037
I:3 Test-Acc:0.0292 Train-Acc:0.04
I:4 Test-Acc:0.0339 Train-Acc:0.046
I:5 Test-Acc:0.0478 Train-Acc:0.068
I:6 Test-Acc:0.076 Train-Acc:0.083
I:7 Test-Acc:0.1316 Train-Acc:0.096
I:8 Test-Acc:0.2137 Train-Acc:0.127

                    ....

I:297 Test-Acc:0.8774 Train-Acc:0.816
I:298 Test-Acc:0.8774 Train-Acc:0.804
I:299 Test-Acc:0.8774 Train-Acc:0.814
```

As you can see, swapping out the first layer from the network in chapter 9 with a convolutional layer gives another few percentage points in error reduction. The output of the convolutional layer (`kernel_output`) is itself also a series of two-dimensional images (the output of each kernel in each input position).

Most uses of convolutional layers stack multiple layers on top of each other, such that each convolutional layer treats the previous as an input image. (Feel free to do this as a personal project; it will increase accuracy further.)

Stacked convolutional layers are one of the main developments that allowed for very deep neural networks (and, by extension, the popularization of the phrase *deep learning*). It can't be overstressed that this invention was a landmark moment for the field; without it, we might still be in the previous AI winter even at the time of writing.

Summary

Reusing weights is one of the most important innovations in deep learning.

Convolutional neural networks are a more general development than you might realize. The notion of reusing weights to increase accuracy is hugely important and has an intuitive basis. Consider what you need to understand in order to detect that a cat is in an image. You first need to understand colors, then lines and edges, corners and small shapes, and eventually the combination of such lower-level features that correspond to a cat. Presumably, neural networks also need to learn about these lower-level features (like lines and edges), and the intelligence for detecting lines and edges is learned in the weights.

But if you use different weights to analyze different parts of an image, each section of weights has to independently learn what a line is. Why? Well, if one set of weights looking at one part of an image learns what a line is, there's no reason to think that another section of weights would somehow have the ability to use that information: it's in a different part of the network.

Convolutions are about taking advantage of a property of learning. Occasionally, you need to use the same idea or piece of intelligence in multiple places; and if that's the case, you should attempt to use the same weights in those locations. This brings us to one of the most important ideas in this book. If you don't learn anything else, learn this:

> **The structure trick**
>
> When a neural network needs to use the same idea in multiple places, endeavor to use the same weights in both places. This will make those weights more intelligent by giving them more samples to learn from, increasing generalization.

Many of the biggest developments in deep learning over the past five years (some before) are iterations of this idea. Convolutions, recurrent neural networks (RNNs), word embeddings, and the recently published capsule networks can all be viewed through this lens. When you know a network will need the same idea in multiple places, force it to use the same weights in those places. I fully expect that more deep learning discoveries will continue to be based on this idea, because it's challenging to discover new, higher-level abstract ideas that neural networks could use repeatedly throughout their architecture.